

EU FREIGHT RATE RISK MONITOR

Air rate risk is 1.6x road, worth EUR 703K a year

Which freight mode carries more rate risk in the EU, and what a buyer does about it

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One contract template, two very different risks

A freight buyer signs air and road contracts the same way. Same twelve-month validity. Same indexation clause. Same tender calendar.

Nobody checks whether the two modes actually behave the same.

If one mode swings much wider than the other, that template is quietly mispriced, and the buyer only finds out at the next surcharge.

What buyers do today

- Price from last quarter's invoices
- Trust the carrier's index story
- Tender when the calendar says, not when the market says
- Hold no reserve for the swing

So I stopped assuming and measured it: two modes, six markets, nineteen years of public price data.

Measure it, do not assume it

Which mode shows more significant rate volatility in the EU, and what seasonal pattern drives it?

6

EU markets
DE ES FR IT NL PL

2

modes
air and road

19

years
2005 Q1 to 2024 Q1

696

market-quarter
observations

Both modes had to exist for the same market and the same quarter, or the comparison is not a comparison.

Public data, tested claims

Where the numbers come from

- Eurostat services producer price index, quarterly, 2021 = 100
- Air NACE H51, road NACE H49.4, five markets
- France is confidential at Eurostat, so France comes from INSEE, freight-only air
- Free open data, no key, fully reproducible

How every claim is tested

- Coefficient of variation: how wide a lane swings for its own level
- Levene test: is the gap between the two modes real, or noise
- Kruskal-Wallis: is the quarter effect real seasonality
- Every chart in the monitor traces back to one of these

Air is the riskier mode, and it is measurable

1.63x

air volatility against road
CV 0.153 vs 0.094

7.0% vs 2.6%

how wide each mode moves
inside a single year

EUR 703K

a year of budget swing
on 10M euro of air spend

The gap is statistically significant, not a story. Levene test p is about 3 in a thousand trillion.

Both modes are seasonal and both peak in the second quarter. Air falls 0.97% in Q1 and climbs 3.20% in Q2.

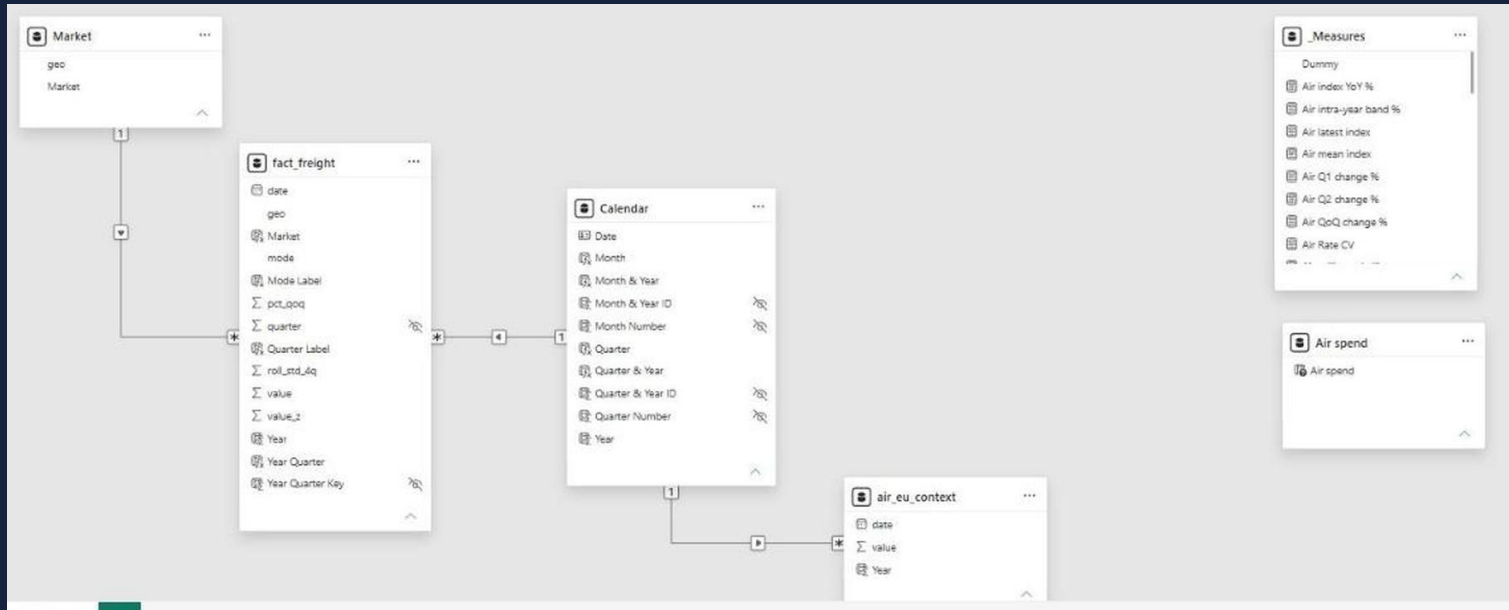
Air CV by market: Italy 0.280 · France 0.185 · Germany 0.146 · Poland 0.119 · Spain 0.107 · Netherlands 0.085

Now the monitor

Ten sheets. Every one answers a single question and ends in an instruction.

Start · Verdict · Markets · History · Timing · Cost · Shocks · Drill · Method · Actions

A star schema, built on purpose



- One fact table in the centre
- A Calendar dimension, marked as the date table, so time intelligence works
- A Market dimension, one hop from the fact
- Fifty DAX measures in their own table
- The two floating tables are deliberate: a measure container and a what-if parameter, which must stay disconnected

Three ways I tried to break my own rule

75%

the trough-lock rule won in
75% of 85 market-years

1.35 - 1.41

the air-road gap holds with
COVID removed entirely

10 of 12

the Q2 climb repeated
out of sample, 2024-2025

A dashboard that recommends is only as good as the rules behind it.

So I backtested the timing rule year by year, re-ran the volatility gap with 2020 to 2022 cut out, and checked whether the pattern continued after the panel ends. It survived all three. Mean edge on the timing rule is 1.8 percent, and in Germany it worked in eight years out of nine.

Where the rule breaks, and what this is not

Where it breaks

- In Poland road is the volatile mode, not air
- In France the timing rule fails, 44 percent hit rate
- Coverage is uneven: Spain 152 quarters, the Netherlands 70
- **So no price hangs on an unvalidated lane**

What this is not

- These are price indices, not the rates you pay
- The Eurostat air index includes passenger air, so it is a broad proxy
- France is a different statistical office, so cross-market levels need care
- **It is a risk monitor, not a forecast**

Four moves a buyer makes on Monday

1

Index-link the air contracts

Air swings 1.6x wider. A fixed air price hands the carrier a free option. Peg it and cap the step.

2

Fix road long

Road CV is 0.094 and the band is 2.6 percent. Lock three years and stop spending negotiation time on it.

3

Tender air into the Q1 trough

Air falls in Q1 and climbs 3.2 percent in Q2. Sign in Q1, start in Q2. Won 75 percent of 85 market-years.

4

Reserve for Italy and the Netherlands

Italy runs CV 0.280 and a 9.58 percent band. The Netherlands is calm on CV but its band is 8.53. Hold cash against both.

What I would say to a buyer in one sentence

Stop pricing air and road the same way.

One of them moves 1.6 times wider, and it costs you EUR 703K a year.

Live monitor and code: <https://rfq.ch/projects/pricing-volatility> · <https://github.com/khareparag/freight-rate-risk-monitor>